

Grade 8
Unit 2
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. Identify the first 10 perfect squares.

$$1^2 = 1 \quad 4^2 = 16 \quad 7^2 = 49 \quad 10^2 = 100$$

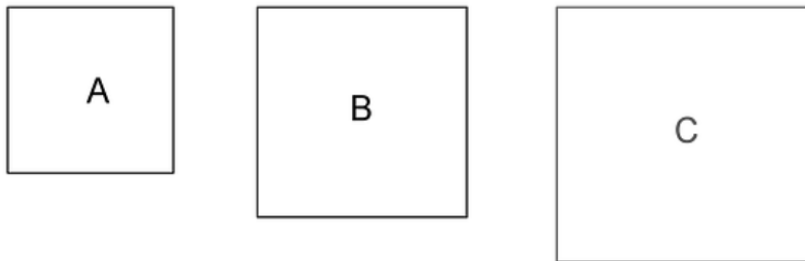
$$2^2 = 4 \quad 5^2 = 25 \quad 8^2 = 64$$

$$3^2 = 9 \quad 6^2 = 36 \quad 9^2 = 81$$

2. Why is 64 a perfect square, but 60 is not?

Because 8 squared gives you 64, but there is no whole number you can multiply by itself that gives you 60.

Square *A* is smaller than Square *B*. Square *B* is smaller than Square *C*.



The three squares' side lengths are $2\frac{3}{5}$ inches, 5 inches, and 7.7 inches. Complete the equation for each of the squares shown.

3. Square *A* Area = $(2\frac{3}{5})^2$ or Area = $2\frac{3}{5} \times 2\frac{3}{5} = 6\frac{19}{25}$ or 6.76 in²

$$2\frac{3}{5} = 2.6 \rightarrow 2.6^2 = 6.76$$

$$2\frac{3}{5} = \frac{13}{5} \rightarrow \left(\frac{13}{5}\right)^2 = \frac{169}{25} = 6\frac{19}{25}$$

4. Square *B* Area = $(5)^2$ or Area = $5 \times 5 = 25$ in²

5. Square *C* Area = $(7.7)^2$ or Area = $7.7 \times 7.7 = 59.29$ in²

Find the length of a side of a square for each area given.

6. $\frac{100}{9}$ square meters

$\frac{10}{3} \cdot \frac{10}{3} = \frac{100}{9}$ side = $\frac{10}{3}$ meters or $3\frac{1}{3}$ or $3.\bar{3}$

7. 2.25 square units

$1.5 \cdot 1.5 = 2.25$ side = 1.5 units

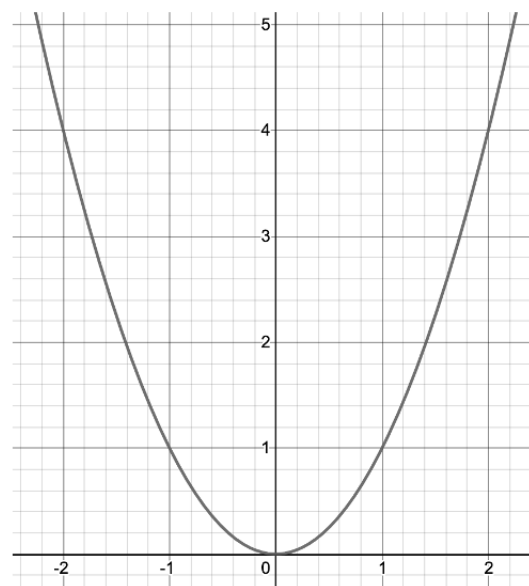
8. 144 square inches

$12 \cdot 12 = 144$ side = 12 inches

The graph represents the equation $y = x^2$.

9. Complete the table of values to find corresponding y -values. Then plot the point on the graph.

x	Substitute into $y = x^2$	Coordinates (x, y)
$-1\frac{3}{5}$	$y = \left(-1\frac{3}{5}\right)^2$ or $y = \left(-\frac{8}{5}\right)^2$	$\left(-1\frac{3}{5}, 2\frac{14}{25}\right)$
-0.4	$y = (-0.4)^2$	$(-0.4, 0.16)$
0	$y = 0^2$	$(0, 0)$
1.2	$y = 1.2^2$	$(1.2, 1.44)$
2	$y = 2^2$	$(2, 4)$



10. Will the graph of $y = x^2$ ever go below the x -axis? Explain why or why not.

No, because any negative value squared is a positive number.